

Lecture 3: Understanding Ethical Problems

HUM 4441: Engineering Ethics

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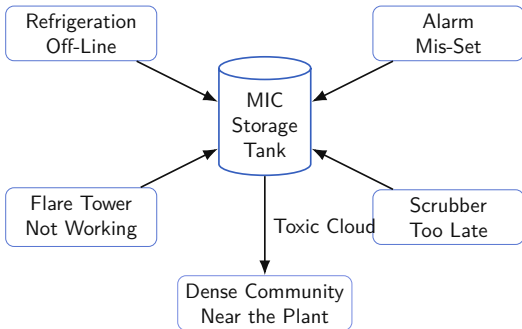
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Bhopal: When a Safety System Becomes a Promise

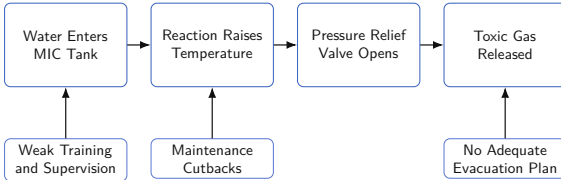


On December 2, 1984, a storage tank at Union Carbide's Bhopal plant released methyl isocyanate gas over nearby neighborhoods. Thousands died, many more were injured, and the harm continued long after the night of the leak.

The ethical problem is not only the chemistry. It is the gap between a dangerous process and the institutions that were supposed to keep people safe.

Case discussed in Fleddermann, Ch. 3

A Disaster Rarely Has One Cause



- The initiating error mattered.
- The disabled safety systems mattered.
- The plant's location near dense settlement mattered.
- The absence of reliable emergency planning mattered.

Which part of this chain looks most like an engineering decision rather than an accident?

Why Ethical Theories Belong in Engineering

In technical courses

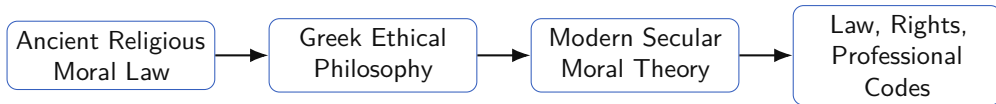
- Concepts define the problem space
- Methods organize messy evidence
- Judgment improves through repeated use

In ethics

- Theories define what counts morally
- Multiple lenses reveal different harms
- Disagreement can be analyzed, not shrugged off

Ethical theory is not decoration. It is a disciplined way to ask what a technical decision does to people.

Where These Ideas Come From



Western ethical theory draws from religious traditions, Greek philosophy, later thinkers such as Locke, Kant, and Mill, and legal codification. For engineering, the important point is practical: ethical conduct is grounded in concern for other people, not merely in law or custom.

Four Questions Behind Four Theories

Utilitarianism

What action produces the best overall consequences for everyone affected?

Duty Ethics

What duties must be honored regardless of convenience or payoff?

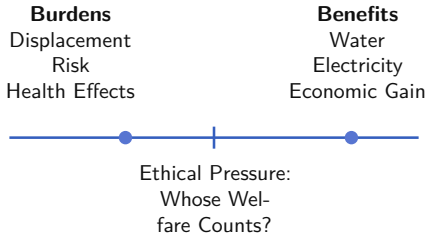
Rights Ethics

Whose rights are at stake, and what would violate them?

Virtue Ethics

What kind of person, team, or profession does this action form?

Utilitarianism: Consequences for Everyone Affected



- Good actions maximize overall well-being.
- The unit of concern is not only the individual decision maker.
- Engineering versions include risk–benefit and cost–benefit reasoning.
- The danger is allowing a small group to carry large costs for everyone else's gain.

A Utilitarian Stress Test

The attraction

- Makes social consequences visible
- Fits many engineering planning tools
- Asks for evidence, not just intuition

Who benefits?

Who pays?

Who is missing from
the calculation?

What if the estimate is wrong?

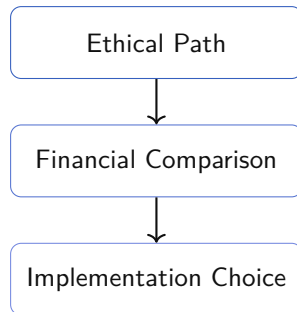
The risk

- Future consequences are uncertain
- Minority harms can look small in the total
- Measurable benefits can overpower invisible losses

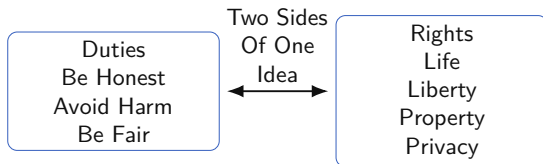
Think of a campus data system. What benefit would be large enough to justify collecting more student data than students expect?

Cost-Benefit Analysis Is Useful, But Not Enough

It can clarify	Direct costs, financial feasibility, expected gains, and trade-offs
It can hide	Dignity, fairness, ecological loss, fear, public trust, and uneven burden
Ethical order	First identify ethical alternatives; then compare their costs



Duty and Rights Ethics: Respect for Persons



- Duties name what professionals owe others.
- Rights name what people may not be forced to surrender.
- Overall benefit matters, but it does not automatically override persons.
- This lens is especially powerful when people are exposed to danger without consent.

When Rights Collide

A public project may protect

- Access to water
- Flood control
- Reliable infrastructure
- Public health

The same project may threaten

- Homes and property
- Local ecosystems
- Community identity
- Meaningful consent

Rights language strengthens ethical analysis, but it does not make conflict disappear.

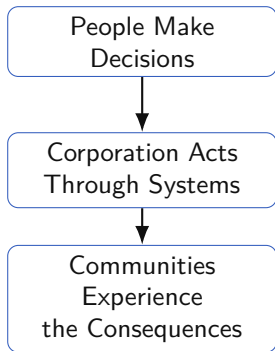
Virtue Ethics: The Character of Practice



Virtue ethics asks what habits, traits, and professional identity a decision expresses.

- Is the action honest?
- Is it responsible toward the people affected?
- Does it show competence or convenient ignorance?
- Would the same action still look honorable if made public?

Can a Corporation Be Moral?



A corporation is not a person in the ordinary moral sense, but corporate action still affects persons. Ethical responsibility cannot be hidden behind an organization chart.

- Organizations should be treated as accountable actors
- Individuals still carry role-based responsibility
- Systems can encourage honesty, silence, diligence, or neglect

Using More Than One Theory

Theory	Strongest question	Watch for
Utilitarianism	What outcome improves overall welfare?	Minority costs disappearing into the total
Duty ethics	What obligations must be honored?	Duties that point in different directions
Rights ethics	Whose rights would be violated?	Unresolved conflicts among rights
Virtue ethics	What character does this action express?	Vague virtue words masking pride or loyalty pressure

When the theories disagree, rights and duties usually deserve especially serious weight because persons should not be treated merely as instruments for aggregate benefit.

Bhopal Through Four Lenses

Utilitarianism

Economic benefit cannot justify exposing a dense community to catastrophic chemical risk without reliable safeguards.

Duty and Rights

The public had a right not to be harmed by preventable industrial danger; the company had duties of maintenance, warning, and emergency planning.

Virtue

Diligence, competence, and honesty are violated when warnings, leaks, and broken safety systems become normal.

Corporate morality

Responsibility spreads across design, management, training, local regulation, and corporate knowledge transfer.

If you were investigating Bhopal, which lens would you start with, and what evidence would you demand first?

Ethics Is Not Geography



The foundations differ across traditions, but the professional results often converge: honesty, protection from harm, fairness, competence, respect, and care for the community.

- Global engineering teams cannot treat ethics as local taste
- Cultural humility matters, but moral relativism is not enough
- Professional codes often converge across countries

Chinese Ethical Traditions

Key emphasis

- Practical moral cultivation
- Ideal character traits
- Mutual respect
- Interdependence of individual and community

Engineering connection

- Integrity and honesty in technical claims
- Design decisions that balance public benefit and individual respect
- Attention to community effects, not only client demands

Indian and Muslim Ethical Traditions

Indian ethics

- Practical formation of a good life
- Virtues such as truth, compassion, self-control, gentleness, and freedom from greed
- Strong fit with professional ideals of service and restraint

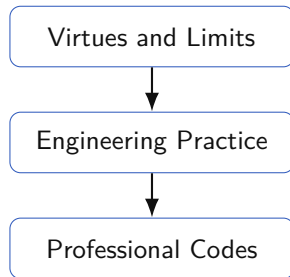
Muslim ethics

- Virtue-oriented moral life grounded in the Qur'an
- Virtues such as humility, honesty, kindness, and trustworthiness
- Professional relevance to truthful claims, fair dealing, and public trust

Buddhist Ethics and Global Codes

Buddhist ethical thinking emphasizes avoiding destruction of life, theft, lying, and harmful excess, while cultivating learning, diligence, generosity, patience, and a sense of limits.

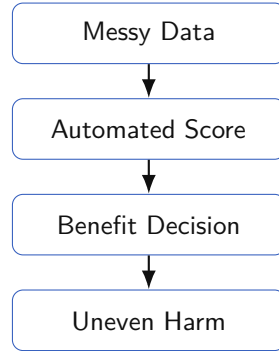
These ideas connect naturally to public safety, environmental responsibility, sustainable development, and continued professional competence.



A Computing Version of the Same Problem

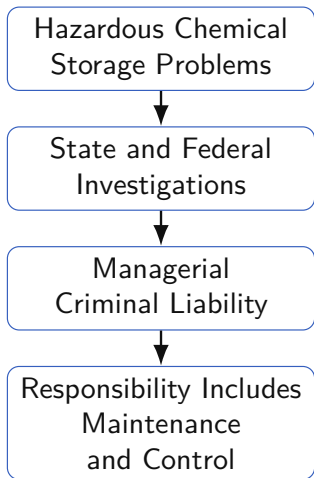
A city wants to deploy an AI system to prioritize public benefits applications. It promises faster decisions and lower administrative cost. Early testing shows higher error rates for applicants whose records are incomplete or inconsistent.

- The benefit is real
- The harm is unevenly distributed
- The public may not understand the system well enough to contest it



Analyze this first as a utilitarian, then as a rights ethicist. What changes?

The Aberdeen Three



Three civilian engineer-managers at the Aberdeen Proving Ground were convicted in 1989 for illegal hazardous-waste storage, treatment, and disposal. They were not necessarily the people who physically handled the chemicals, but they were responsible for the conditions under their authority.

Case discussed in Fleddermann, Ch. 3

Teamwork Has an Ethical Dimension

Failures that hurt the team

- Not completing assigned work
- Hiding delays or errors
- Withholding information
- Taking over so others cannot contribute

Virtues that support the team

- Reliability
- Openness
- Fairness
- Shared responsibility

In a software project, when does “I will just do it myself” become an ethical problem rather than efficiency?

Lecture 3 Summary

- Ethical theories give engineers disciplined ways to analyze harm, benefit, duty, rights, character, and responsibility.
- Utilitarianism and cost–benefit reasoning are powerful, but they can hide uneven burdens and hard-to-price losses.
- Duty and rights ethics protect persons from being treated only as means to an aggregate goal.
- Virtue ethics asks what kind of professional character a decision expresses and reinforces.
- Bhopal shows why ethical analysis must include systems, maintenance, warnings, emergency planning, corporate accountability, and local context.
- Ethical traditions differ in origin, but engineering practice converges around honesty, competence, safety, fairness, and respect for people.

References

- Charles B. Fleddermann, *Engineering Ethics*, 4th ed., Chapter 3.
- Charles E. Harris Jr., Michael S. Pritchard, and Michael J. Rabins, *Engineering Ethics: Concepts and Cases*.
- Roland Schinzinger and Mike W. Martin, *Introduction to Engineering Ethics*.